Molecular Dynamics Modeling: Evaluation of Grain Boundary Energies and Fracture in Σ5(310) Aluminum

Shira Rys Corpuz[1], Wanjuan Zou[2], Penghui Cao[2,1]

[1] Department of Materials Science and Engineering, University of California, Irvine, [2] Department of Mechanical and Aerospace Engineering, University of California, Irvine

Introduction

Material failure, such as fracture, fatigue, and corrosion can compromise safety and reliability in engineering structures. Experimental tests (tensile, fatigue, impact) provide valuable insights but are often costly and time-consuming. In contrast, Molecular Dynamics (MD) simulations is a computational method that uses Newton's laws of motion to numerically integrate atomic forces and trajectories, offering atomic-level insight into failure mechanisms.[1]

Among various microstructural features influencing material failure, grain boundaries (GBs) play a pivotal role. For example, GBs are regions of crystallographic mismatch that act as preferred sites for weakening agents due to their higher energy and atomic disorder compared to the grain interior.[2] Understanding the atomic structure of GBs is therefore essential.

In this work, a Σ5(310) symmetric tilt grain boundary in aluminum was constructed using MD simulations in LAMMPS, and the GB energy was also computed. Preliminary tensile tests were also performed. By focusing on aluminum, a well-characterized FCC metal, we aim to better understand GB behavior before extending the study to more complex materials such as high-entropy alloys.

Background

Total Potential Energy — Cohesive Energy

$$\gamma_{GB} = \frac{\boxed{E_{GB}} - \boxed{N} \cdot \boxed{E_{coh}}}{2\boxed{A}}$$

\# of Atoms — GB Area

Figure (Above): Grain Boundary Energy Equation

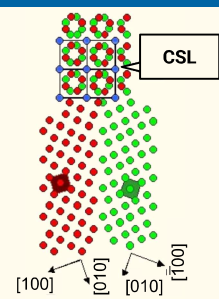


Figure (Right): Coincidence Site Lattice (CSL)[3]

Method

Grain Boundary Generation

1	LAMMPS Code	Made to extract • cohesive energy • lattice constants • grain boundary structures
2	Set Up Lattice and Box	Lattice constant: 4.05 Å Region dimensions: • X = 0 Å to 12.81 Å • Y = -64.04 Å to 64.04 Å • Z = 0 Å to 4.050 Å
3	Create Symmetric Tilt GB	Two blocks of the atomic structure of Al Σ5(310) was created to make a symmetric tilt grain boundary
4	Equilibrate the System at 0K	Define Interatomic Potential with Aluminum potential[4] Run Minimization to relax 1. atoms 2. atoms and box to 0 pressure
5	Compute GB Energy	LAMMPS computes this with the GB energy equation
6	Generate Visuals	Take Dump File into OVITO for visual processing

Fracture Generation

1	LAMMPS Code	Made to: • Apply 0.1% per loop incremental tensile strain along the y-axis • Delete atoms beyond 50 Å from GB • Fix atoms within 45 Å from GB
2	Input GB Datafile	Prompt code to use created GB datafile
3	Generate Visuals	Place Dump File into OVITO for visual processing

Results

Grain Boundary

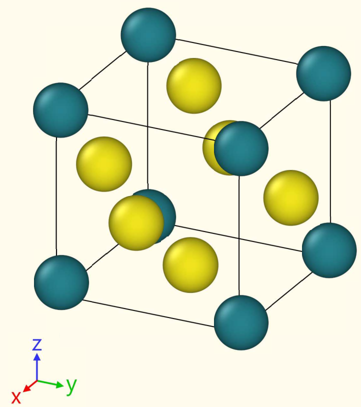
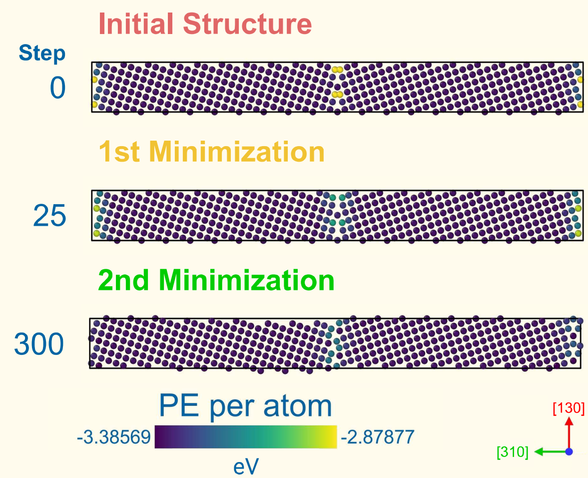


Figure: Face Centered Cubic (FCC) Crystal Structure Face (Yellow), Teal (Corner)



Recorded Energies

Grain Boundary Energy: 564 mJ/m^2

Fracture via Modeled Tensile Test

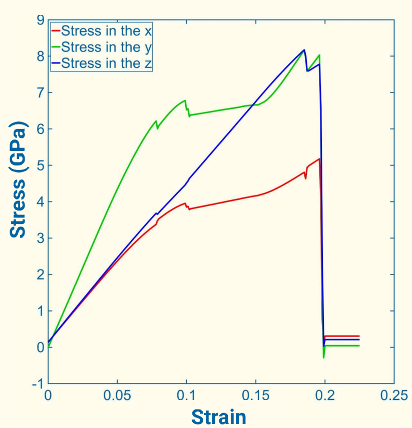
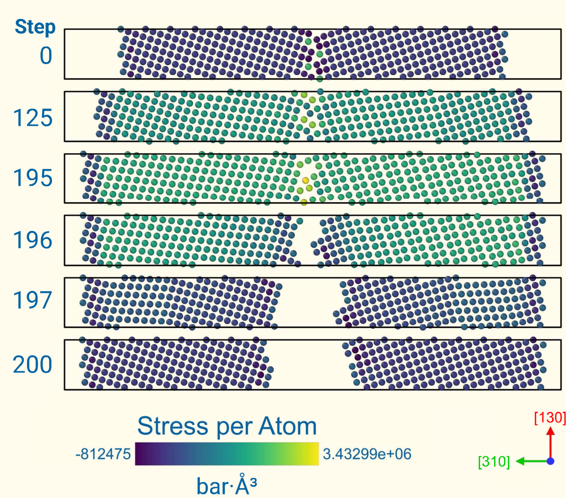


Figure: Stress-Strain Curve from of Simulated Tensile Test



Conclusions

The calculated grain boundary energy of 564 mJ/m² falls within the 500–600 mJ/m² theoretical range for Σ5(310) aluminum reported in experimental literature.[5] The fracture model behaved predictably, consistent with established theory.[6]

These simulations help one understand Molecular Dynamics (MD) as an inexpensive, reliable method for investigating material behavior. The approach can be extended to more complex systems such as high-entropy alloys (HEAs), with machine learning offering potential to enhance the modeling process.

References

1. Zou W. 2023 Jul 24. *An Introduction to Molecular Dynamics (MD) Simulation* [PowerPoint presentation]. Irvine (CA): Prof. Penghui Cao's Lab, University of California, Irvine; Materials REU 2023.
2. Gottstein G. *Grain Boundary Migration in Metals: Thermodynamics, Kinetics, Applications*. 2nd ed. Boca Raton (FL): CRC Press; 2021. p 115-130.
3. Trinkle DR. *7.1.2 The Coincidence Site Lattice* [Internet]. Urbana (IL): University of Illinois at Urbana–Champaign; c2023 [cited 2025 Aug 11]. Available from: https://dtrinkle.matse.illinois.edu/MatSE584/kap_7/backbone/r7_1_2.html
4. Mishin Y, Farkas D, Mehl MJ, Papaconstantopoulos DA. *Interatomic Potentials for Monoatomic Metals from Experimental Data and ab Initio Calculations. Phys Rev B*. 1999;59(5):3393-3407. Available from: http://www.ctcms.nist.gov/potentials
5. Tschopp MR. *LAMMPS Tutorials* [Internet]. GitHub; c2023 [cited 2025 Aug 11]. Available from: https://github.com/mrkllntschpp/lammps-tutorials
6. Callister WD, Rethwisch DG. *Materials Science and Engineering: An Introduction*. 10th ed. Hoboken (NJ): Wiley; 2018. p 142-165.

Acknowledgements

Special thanks to Wanjuan Zou, Professor Penghui Cao, Cao Research Group, Denise Ho, and the 2025 CCAM cohort

Center for Complex and Active Materials
UCI MRSEC

UC IRVINE MATERIALS RESEARCH INSTITUTE

National Science Foundation
MRSEC
Materials Research Science and Engineering Centers